PRISM Energy Intelligence

Operating updates need financial context.

Analysts need a traceable way to test early operating signals
before using them as investment evidence.

What I built	An NTPC dashboard with a generation backtest, financial assessment and Prismy.
Who it helps	Prism’s research analysts and product team.
Why Prism	Financial and alternative-data research with transparent calculations and source-linked answers.

How the project answers the task

Task requirement	Project response
Define the problem	Help Prism analysts test operating signals and assess their financial meaning.
Build a prototype	Public CEA and NTPC data, transparent calculations and a working dashboard.
Present the result	A source-backed investment brief: watchlist assessment, cash risks and an editable price screen.
Document the work	Code, README, source files, 28 tests, assumptions, limitations and next steps.

Main deliverable: Investment decision. Supporting evidence: the historical evaluation and original sources.

Project flowchart

Operating evidence

CEA reports
Same eight stations, originals and hashes

Historical cutoff and quarter output
Eligible reports only; fiscal YTD becomes QTD

Forecast and evaluate
Run rate, two baselines; later actuals score only

Financial evidence

NTPC filings and NSE close
Consolidated results and a dated market price

Earnings, cash and debt
Tax effects, capex and financing interest

Valuation assumptions
Earnings haircut, fair P/E and safety margin

Investment brief: operating trend, financial quality and price discipline

Prismy Matches questions to fixed evidence cards and source links.

The useful output: an investment research brief

Watchlist

What the evidence says

Q1 FY27 owners' profit grew 11.8% YoY.

Tax and regulatory effects complicate recurring earnings.

Cash after capex does not cover cash interest.

Example price screen

₹303.75

28 Oct 2026 close; reported P/E 10.61×

₹248.22

Example entry threshold

15% earnings haircut, 12× fair P/E, 15% safety margin. Editable assumptions.

A stronger buy case needs recurring earnings, returns from new capacity and a credible funding plan. No return forecast is established.

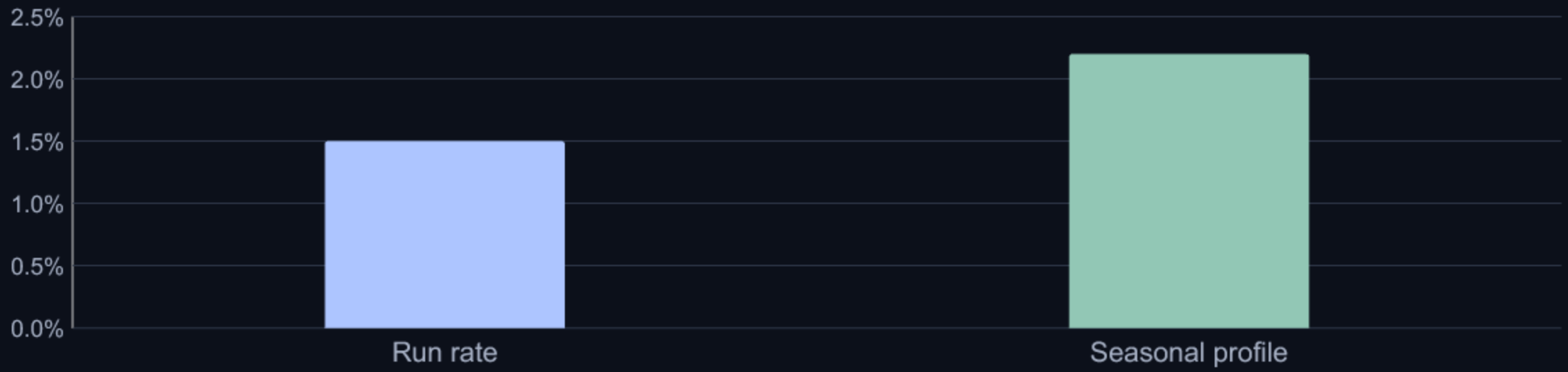
The generation model adds useful, limited evidence

1.50%

run-rate average error

2.20%

seasonal-profile average error

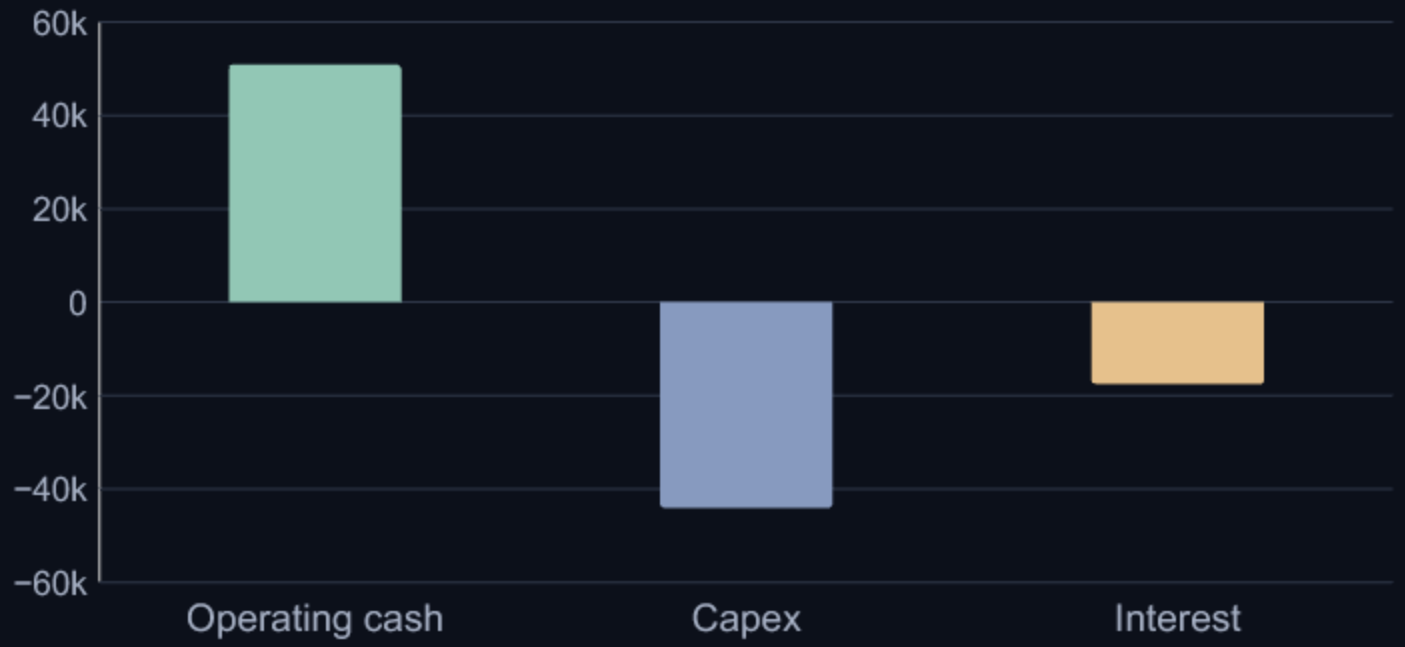


Run rate wins 5 of 8 quarters. Seasonal profile wins 3.

Remaining-output error: 3.81% vs 5.54%. Publication proxies and the small sample limit confidence.

Cash interest changes the financial interpretation

FY26 cash flows, ₹ crore



Operating cash

₹50,901.81 cr

Cash capital spending

₹44,049.78 cr

Cash interest

₹17,537.73 cr

After capex: +₹6,852 cr. After cash interest: −₹10,686 cr.

NTPC classifies interest in financing. This selected bridge excludes other investing and financing flows.

What I noticed, and what I would improve

Observation	Design choice or next improvement
Most of the quarter is already known	Show remaining-output error alongside full-quarter MAPE.
Publication times are unverified	Use explicit proxies now; capture first-public timestamps and vintages next.
Tax and regulatory effects interact	Flag earnings quality; build recurring-earnings and funding scenarios next.
Free-form answers can introduce claims	Prismy renders canonical evidence cards with sources. Expand relevance testing next.

Broader station coverage and prospective holdout quarters would strengthen the operating study.

The project is ready to review

Live dashboard

https://ronnismad.github.io/prism-ntpc-research-dashboard/

Code, data and README

https://github.com/ronnismad/prism-ntpc-research-dashboard

Suggested review

Investment decision, Historical evaluation, Methodology, then a Prismy question.

Reproduce locally

npm install, npm test, npm run dev. The source includes 28 tests and preserved reports.

Prismy on the hosted demo uses instant matching. Local Ollama is an optional mode in the source.